Data Structures II

CPSC 482

Dr. Hossain

DSM Tutorial
Compiling and Running

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Compiling and Running

Prerequisites

- C++17 compiler (`clang++` or `g++`)
- GNU Make
- Python 3 with packages: `numpy`, `pandas`, `matplotlib`

Compiling

From the project root directory:

```
mkdir out && make
```

This produces the executable `out/main`. To clean build artifacts:

```
make clean
```

Running – Interactive Mode

Run with no arguments to process a single matrix interactively:

```
./out/main
```

You will be prompted to select a `.mtx` file from `data/`. The program runs the full optimization pipeline and prints the before/after metrics to the terminal.

Running – Benchmark Mode

Run with the `-bench` flag to process all benchmark matrices with timing:

```
./out/main --bench
```

This generates CSV results and permuted matrices under `out/`.

Generating Plots

After running the benchmark, you can run the plotting scripts:

```
python3 scripts/plot_benchmarks.py
python3 scripts/plot_optimization_metrics.py
python3 scripts/generate_report_tables.py
python3 scripts/plot_reordering.py
```

These read the CSV files from `out/` and produce figures and tables, also saved under `out/`.